

regardless of the sign of n , this condition is written as

$$e^2 < 2\mu(|n| - a). \quad (102)$$

Let us now consider the second case $|n| < a$. For this case, the singularity is represented by a part of the circle (101) such that the point with $r_s = D$ must be included, since the interval $[x_{\min}, x_{\max}]$ will contain the point $x = 0$ (Fig. 1). Thus, the largest value of the radial coordinate corresponding to the singularity is $r_s = D$, and in order for the solution not to be a naked singularity, we must require the condition $r_H^+ > r_s$, which can be written using the relation between the dilaton and the electromagnetic charge as

$$\sqrt{(e^2 - 2\mu^2)^2 - 4a^2\mu^2} > e^2 - 2m\mu. \quad (103)$$

If we consider the case $e^2 > 2m\mu$ then the square of this inequality leads to

$$(\mu - m)e^2 + (a^2 - n^2)\mu < 0, \quad (104)$$

which cannot be satisfied due to $|n| < a$, so we have to consider $e^2 < 2m\mu$, which satisfies the inequality (103) by default. One more constraint following from the existence of the horizon is

$$e^2 < 2\mu(\mu - a). \quad (105)$$

One obtains, that in the case $|n| < a$ the relation $\mu - a < m$ must be fulfilled, so the condition for the existence of the horizon is enough for (103) to be satisfied. Than in order for electromagnetic charge e to remain real one also has to impose the condition $\mu - a > 0$, which leads to

$$m > \sqrt{a^2 - n^2}. \quad (106)$$

So in the case $|n| < a$ the conditions (105) and (106) guarantee the absence of a naked singularity.

VI. GAUGED EMDA

Now we come back to the full theory with a potential as described in Section 3.

A. Forth order polynomials for A_2 , B_2

In the gauged case, the functions A_{23} , B_{23} , R , Y can be considered exactly the same as in the previous section (70), and axidilaton will also take the general form (57). But we need to